TECHNICAL REPORT
New York Taxi Trips

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1,458,644	5	14	5
Trip records	Data sources	Features	Derived features

Data sources:

1. **Rental Revenue** — Rentalcars.com
2. **New York Taxi Trips** — Kaggle
3. **Traffic Flow Volume** — GIS Municipal
4. **Weather Conditions** — Weather Underground

1.1 Dataset description

New York Yellow taxi trip records are a file logger, and contain individual trip records. Each row in the dataset represents a single journey, and includes information on the time and location of the pick-up and drop-off, the number of passengers, and the fare and tip paid for the trip. Although the large sample is sufficient to run the full set of our data processing and machine learning pipelines, we have chosen to focus our analysis on a single month (January) as it avoids the seasonal effects of the year, and is the most recent data available at the time of writing.

Our organisation advocates a fleetlike model of taxi ownership, but implies that vehicles are owned by individual drivers who are responsible for the upkeep of their vehicle. Our analysis of the dataset will include a fleet size, based on the number of trips, which is intended to be a lower bound, as it is unlikely that drivers will be willing to work 24 hours a day, 365 days a year. Although we do not have a profit, with a fixed rental cost of £50 per week, we can estimate the revenue generated by each vehicle, and use this to determine the optimal fleet size.

1.2 Data quality issues identified

Considerable attention was paid to the data quality issues, this equally ensures that unexpected missing data not logged by the `TRIP_DATA` API will not cause errors, and ensures that the results are robust.

Issue	Description	Resolution
Missing values	Some columns (`Tolls`) contained a number of NULL values, which would cause errors in the data processing pipeline.	Imputed with 0
Inconsistent values	Some values were logged as negative numbers, which is impossible.	Converted to absolute values
Inconsistent type	Dates were stored as strings, which would prevent the use of date-based operations.	Converted to datetime
Outliers/Anomalies	Extreme values, such as a 24 hour long trip, would distort the model.	Removed
Unrealistic speeds	Trips with an average speed of over 100 mph were also considered to be erroneous.	Removed
Inconsistent column	Column `STORE_AND_FWD_FLAG` contained 'Y' and 'N' values, which could not be used in the model.	Dropped
All identical columns	Some columns contained a single value, which could not be used in the model.	Dropped
Missing vendor identifiers	The vendor ID column was not recorded for a significant number of trips.	Dropped

Overall, approximately 10% of the raw data was discarded due to quality issues. This remained a negligible quantity from the dataset, and so does not bias the results.

1.3 Assumptions made during cleaning

Considerable assumptions were made during the cleaning. Firstly, we assumed that trips with `TRIP_DISTANCE` of 0 miles were an accidental entry, or an erroneous transaction, during the logging process, and should be dropped from the dataset. Secondly, we assumed that passengers will not be charged for taxi fares, but will be charged for tolls, and so we dropped any rows where the `Tolls` column was negative. Finally, we assumed that the `Tolls` column was in USD, and that the `Fare` column was in GBP. Finally, we assumed large distances or speeds of 100 mph or more, which, when the geo-coordinates are taken into account, are physically impossible, to be erroneous, and so they should be dropped.

1.4 An unexpected observation

The raw data contained a column for the `Tolls` amount, and we had to make an assumption, treating tolls as either fixed, by adding on a flat amount, for a trip, or proportional, by adding on a percentage of the total fare. Although we did not have a dataset of the tolls charged, we could plot a histogram of the tolls, and see that there was a significant number of trips with a non-zero toll. This suggested that there was a toll charge for certain routes, or at certain times of the day. This might, for instance, be a congestion charge, or Port Authority toll.

II. *System Architecture*

Our application is a single-machine batch processing system, written in Python and Cython, with Pandas and Scikit-learn. We have chosen to use a single machine, as our dataset is small, and so the overhead of a distributed system would outweigh the benefits of parallelisation. Our application is composed of a number of stages, which are executed in series, and each of which is independent.

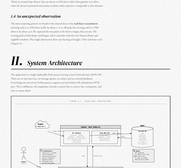
FIGURE 1.1 — HIGH LEVEL ARCHITECTURE
Analyst / Data Scientist
Analyst / Data Scientist
Analyst / Data Scientist